SIMATS ENGINEERING
CO3-ASSSESSMENT TOOL-2
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Course Code: **CSA1511** **Slot:** B

Course Name: **CLOUD COMPUTING AND BIG DATA ANALYTICS**

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Case Study 1: Smart Retail Inventory Management System

1. Cloud Computing Model Used

The system uses a combination of **IaaS, PaaS, and SaaS**.

- **IaaS:** Virtual machines, networking, VPNs and firewalls.
- **PaaS:** Cloud platform services for analytics and demand forecasting.
- **SaaS:** Web-based inventory dashboards and applications.

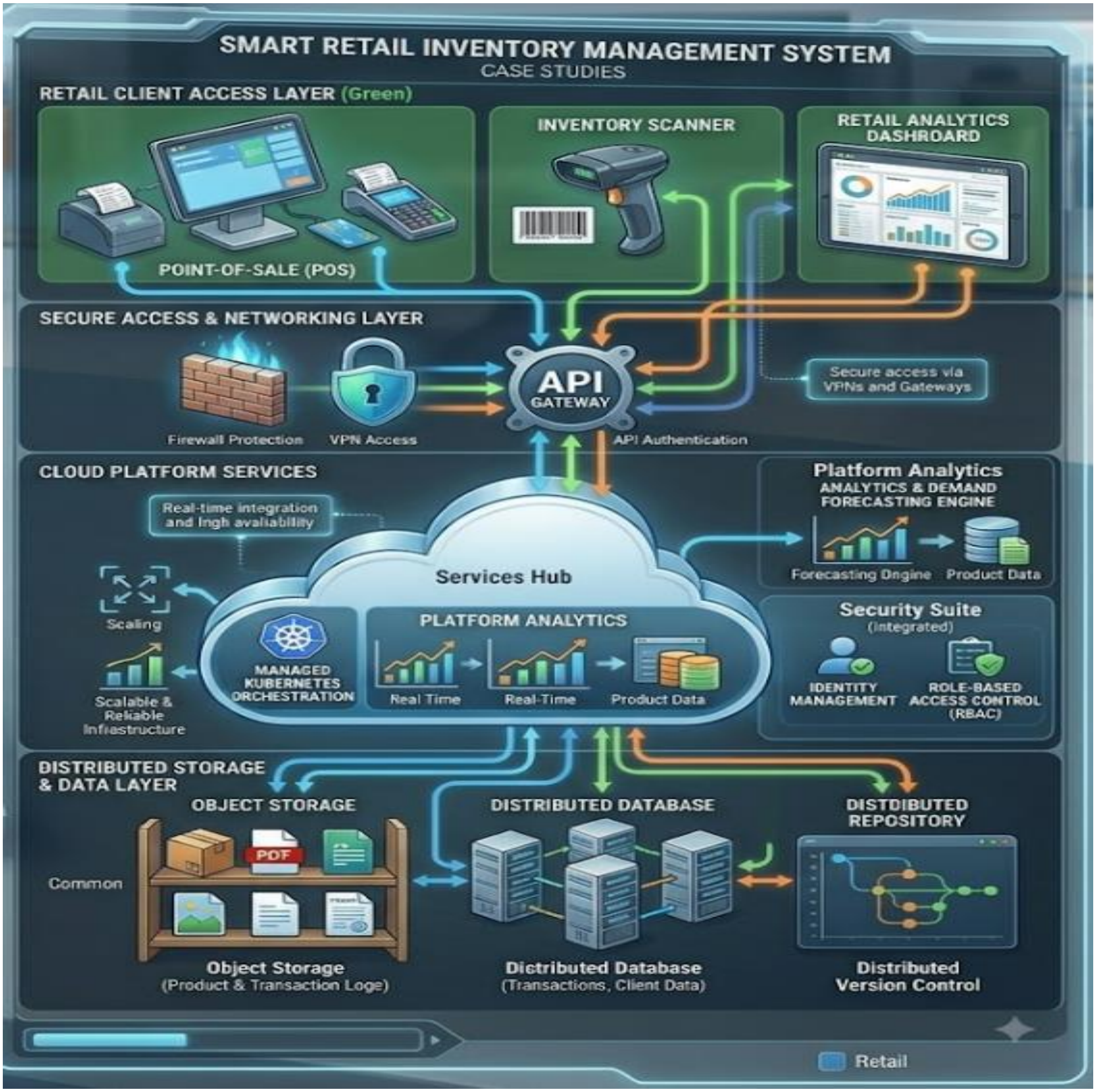
2. Main Components

1. POS systems
2. Web dashboards
3. APIs
4. API Gateway
5. Cloud storage
6. Analytics services
7. Virtual machines
8. VPN
9. Firewall

3. Working

POS Systems → APIs/API Gateway → Cloud Services → Cloud Storage → Analytics → Dashboard

POS systems collect sales information. The data is sent through APIs to centralized cloud services. Cloud storage stores product information and transaction logs. Platform services analyze the data and help with demand forecasting. Dashboards display inventory and analytics information.



4. Security

- VPN
- Firewalls
- API Gateway
- Authentication
- Access control
- Secure communication

5. Benefits

- Centralized inventory management
- Real-time data access
- Better demand forecasting
- Improved scalability
- Secure data storage
- Reduced infrastructure maintenance

Conclusion

The Smart Retail Inventory Management System uses cloud infrastructure, APIs, storage, analytics and security services to provide **efficient, scalable and secure inventory management**.

Case Study 2: Cloud-Based Document Collaboration System

1. Cloud Computing Model Used

The system mainly uses **SaaS**, supported by PaaS and IaaS.

- **IaaS:** Cloud servers and network infrastructure.
- **PaaS:** Platform services for synchronization and application development.
- **SaaS:** Online document collaboration application.

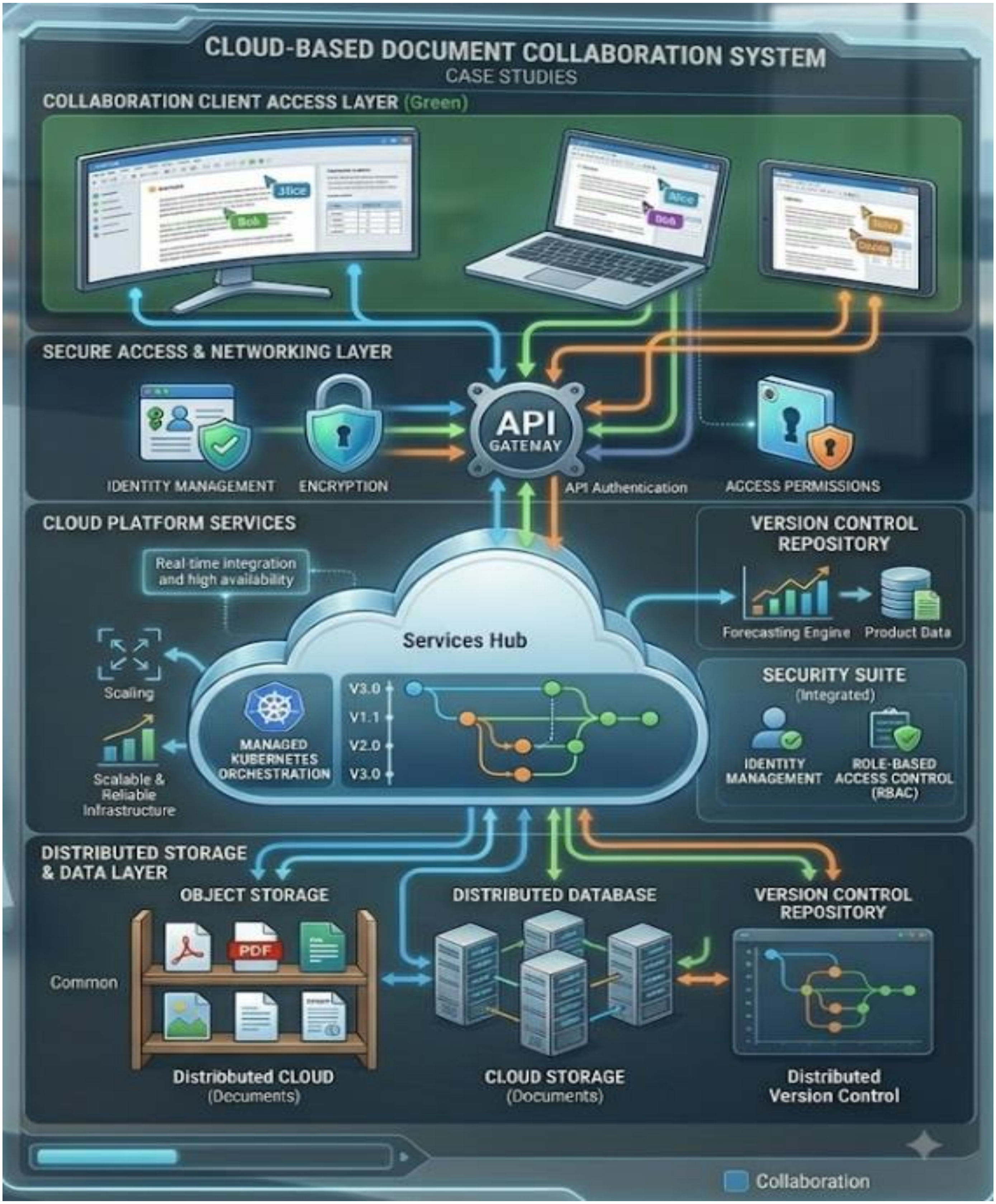
2. Main Components

1. Web browsers
2. Document collaboration application
3. Distributed cloud storage
4. APIs
5. Version control
6. Network infrastructure
7. Identity management
8. Access control
9. Encryption

3. Working

User → Web Browser → Cloud Application → APIs → Cloud Storage

Users access the application through web browsers. Documents are stored in distributed cloud storage. APIs synchronize documents between different users and devices. Version control maintains different versions of documents.



CLOUD-BASED DOCUMENT COLLABORATION SYSTEM
CASE STUDIES
COLLABORATION CLIENT ACCESS LAYER (Green)
Alice
Bob
SECURE ACCESS & NETWORKING LAYER
API
GATEWAY
IDENTITY MANAGEMENT
ENCRYPTION
API Authentication
ACCESS PERMISSIONS
CLOUD PLATFORM SERVICES
Real time integration and high availability
Services Hub
Scaling
Scalable & Reliable Infrastructure
MANAGED KUBERNETES ORCHESTRATION
V3.0
V1.1
V2.0
V3.0
VERSION CONTROL REPOSITORY
Forecasting Engine
Product Data
SECURITY SUITE
(Integrated)
IDENTITY MANAGEMENT
ROLE-BASED ACCESS CONTROL (RBAC)
DISTRIBUTED STORAGE & DATA LAYER
OBJECT STORAGE
DISTRIBUTED DATABASE
VERSION CONTROL REPOSITORY
PDF
Common
Distributed CLOUD
(Documents)
CLOUD STORAGE
(Documents)
Distributed
Version Control
Collaboration

4. Security

- Encryption
- Authentication
- Identity management
- Access permissions
- Role-based access control
- Secure network communication

5. Benefits

- Real-time collaboration
- Multiple users can edit documents
- Access from different devices
- Automatic synchronization
- Version control
- High availability
- Global accessibility

Conclusion

The Cloud-Based Document Collaboration System provides **real-time document editing, sharing and synchronization** using cloud computing and secure network services.

Case Study 3: Online Food Ordering Platform

1. Cloud Computing Model Used

The system uses **IaaS, PaaS and SaaS**.

- **IaaS:** Virtual machines and cloud infrastructure.
- **PaaS:** Managed Kubernetes and cloud platform services.
- **SaaS:** Online food ordering application.

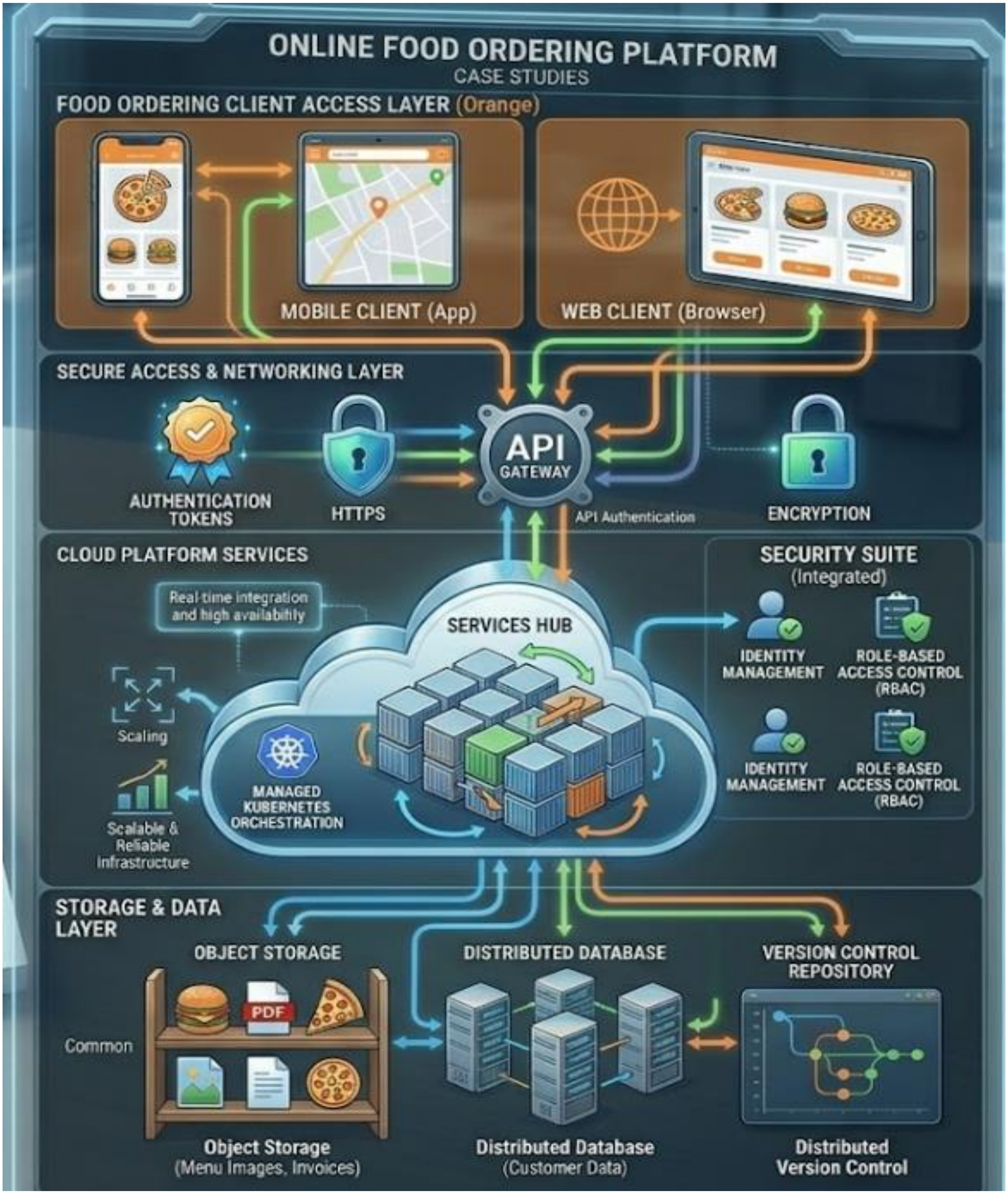
2. Main Components

1. Web browser
2. Mobile application
3. RESTful Web APIs
4. Backend services
5. Virtual machines
6. Managed Kubernetes
7. Cloud storage
8. Authentication service
9. HTTPS
10. Role-Based Access Control

3. Working

Customer → Web/Mobile App → REST API → Backend Services → Kubernetes/VMs → Cloud Storage

Customers access the platform through a browser or mobile application. The frontend communicates with backend services using RESTful APIs. Backend applications run on virtual machines and managed Kubernetes. Cloud storage stores menu images, invoices and customer information.


ONLINE FOOD ORDERING PLATFORM
CASE STUDIES
FOOD ORDERING CLIENT ACCESS LAYER (Orange)
MOBILE CLIENT (App)
WEB CLIENT (Browser)
SECURE ACCESS & NETWORKING LAYER
AUTHENTICATION TOKENS
HTTPS
API GATEWAY
API Authentication
ENCRYPTION
CLOUD PLATFORM SERVICES
Real-time integration and high availability
SERVICES HUB
Scaling
MANAGED KUBERNETES ORCHESTRATION
Scalable & Reliable Infrastructure
SECURITY SUITE
(Integrated)
IDENTITY MANAGEMENT
ROLE-BASED ACCESS CONTROL (RBAC)
IDENTITY MANAGEMENT
ROLE-BASED ACCESS CONTROL (RBAC)
STORAGE & DATA LAYER
OBJECT STORAGE
DISTRIBUTED DATABASE
VERSION CONTROL REPOSITORY
Common
PDF
Object Storage
(Menu Images, Invoices)
Distributed Database
(Customer Data)
Distributed Version Control

4. Security

- Authentication tokens
- HTTPS
- Role-Based Access Control (RBAC)
- User authentication
- Authorization
- Secure cloud storage

5. Benefits

- Online food ordering
- Web and mobile access
- High scalability
- Reliable application performance
- Secure customer data
- Easy management of large numbers of orders
- High availability

Conclusion

The Online Food Ordering Platform demonstrates the integration of **clients, REST APIs, cloud storage, virtual machines, Kubernetes and security services** to create a scalable and reliable cloud application.